Prompted Segmentation for Drywall QA

Take-Home Assignment | AI Research Intern | Origin (formerly 10xConstruction)

Pranav Prashant Shewale | Seed: 42 | Device: CUDA (Colab T4)

Model	CLIPSeg (CIDAS/clipseg-rd64-refined)
Task	Text-conditioned binary segmentation
Prompts	"segment taping area" | "segment crack"
Taping mIoU / Dice	0.6051 / 0.7401
Crack mIoU / Dice	0.5178 / 0.6646
Train Time	63.7 min (10 epochs, Colab T4)
Avg Inference	36.2 ms / image (GPU)
Checkpoint Size	603.2 MB

1. Methodology

Model Architecture

CLIPSeg (*CIDAS/clipseg-rd64-refined*) extends CLIP with a lightweight FiLM-conditioned transformer decoder to produce dense binary segmentation maps from free-text prompts. Given an image and a natural-language query, it fuses CLIP's shared vision–language embedding with a convolutional decoder to output per-pixel logits at reduced resolution (64px patches), then upsampled to the original image size via bilinear interpolation.

Why CLIPSeg over GroundedSAM?

Criterion	CLIPSeg	GroundedSAM
Text-conditioned	Native — single forward pass	Requires separate Grounding-DINO stage
GPU memory (T4)	~2 GB	~8+ GB
Fine-tunable	Yes — decoder only (~0.5M params)	Complex multi-model pipeline
Inference speed	~36 ms / image	~200+ ms / image

Training Strategy

- **Backbone frozen:** All CLIP encoder weights are frozen. Only the CLIPSeg FiLM decoder (~0.5M params) is fine-tuned, preserving rich visual-language alignment and preventing overfitting on small datasets.
- **Loss — BCE + Dice (50/50):** BCE provides pixel-level calibration; Dice directly optimises overlap and combats class imbalance inherent in thin seams and hairline cracks.
- **Prompt augmentation:** 4 phrasings per class randomly sampled during training. Canonical prompt used at inference for reproducibility and consistency.
- **Optimiser:** AdamW (lr=1e-4, weight_decay=1e-4) with CosineAnnealingLR scheduler over 10 epochs (eta_min=1e-6).
- **Gradient clipping:** Max norm = 1.0 to stabilise the mixed-dataset training loader.

Hyperparameters

Parameter	Value
Epochs	10
Batch size	4
Learning rate	1e-4 (AdamW, weight_decay=1e-4)
LR schedule	CosineAnnealingLR (T_max=10, eta_min=1e-6)
Image size	352 × 352
Loss weights	BCE 0.5 | Dice 0.5

Seed	42

2. Data Preparation

Datasets

Both datasets were sourced from Roboflow Universe and forked into a personal Roboflow workspace (*general-okp9d*). A dataset version was then generated for each project to enable programmatic download via the Roboflow Python API. The annotations (bounding boxes) were already present in the original public datasets — no manual labelling was performed.

	Dataset 1 — Taping Area	Dataset 2 — Cracks
Source	Roboflow: drywall-join-detect	Roboflow: cracks-3ii36
Project slug	drywall-join-detect-vj9mn	cracks-3ii36-wbdmt
Annotation	COCO bbox (pre-existing)	COCO bbox (pre-existing)
Classes	drywall-joint	crack
Total images	1,022	5,369
Train	820	5,164
Val	202	201
Test	—	4
Prompt	"segment taping area"	"segment crack"

Mask Generation Pipeline

The Roboflow COCO export provides bounding box annotations. These were converted to binary mask PNGs required for segmentation training via the following pipeline:

- **Step 1 — Download:** Datasets downloaded via Roboflow Python API (COCO format) from the forked workspace versions.
- **Step 2 — Parse COCO JSON:** Load *_annotations.coco.json* for each split (train/valid/test).
- **Step 3 — Polygon fill (primary):** If polygon *segmentation* fields exist, fill with *cv2.fillPoly*.
- **Step 4 — Bbox fallback:** If only bounding box annotations are present, fill rectangle with *cv2.rectangle* to create an approximate foreground mask.
- **Step 5 — Binarise:** All non-zero mask pixels set to 255. Output: single-channel PNG, values {0, 255}.
- **Step 6 — Resize:** Images and masks both resized to 352×352 (CLIPSeg native resolution) via bilinear / nearest-neighbour interpolation respectively.

Prompt Augmentation

Class	Training Prompts (randomly sampled per image)	Inference Prompt
Taping	"segment taping area", "segment joint tape", "segment drywall seam", "segment wall joint"	"segment taping area"

Cracks	"segment crack", "segment wall crack", "segment surface crack", "segment fracture"	"segment crack"

3. Results

Quantitative Metrics

Evaluated on the held-out validation set using the best checkpoint, selected by average mIoU across both classes:

Metric	Taping Area (segment taping area)	Cracks (segment crack)
mIoU	0.6051	0.5178
Dice	0.7401	0.6646
Precision	0.7385	0.6851
Recall	0.7770	0.7261

Runtime & Model Footprint

Metric	Value
Total training time	63.7 min (10 epochs, Colab T4 GPU)
Avg inference time	36.2 ms / image (GPU)
Model checkpoint size	603.2 MB
Total parameters	~67M
Trainable parameters	~0.5M (decoder only — backbone frozen)

Training Curves

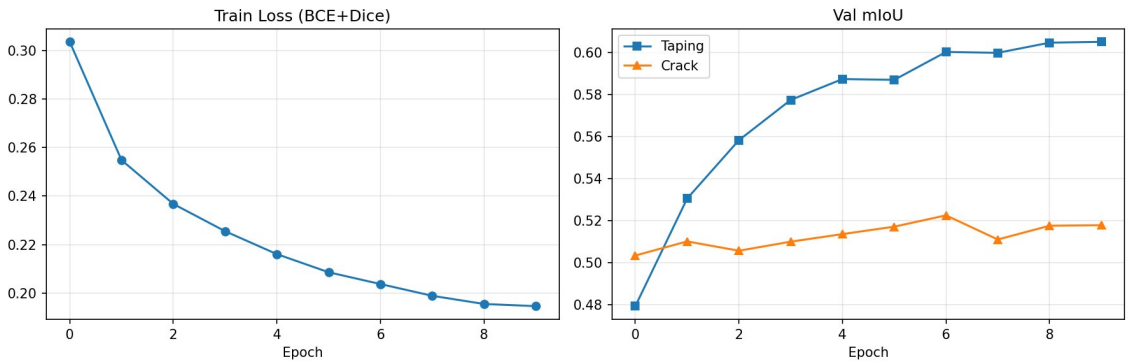

Figure 1: Training loss (BCE+Dice) and validation mIoU across 10 epochs. Loss converges smoothly; taping plateaus at ~0.60, crack at ~0.52.

Qualitative Results — Taping Area

Input image | Ground truth mask | Predicted mask. Per-image IoU shown in title.

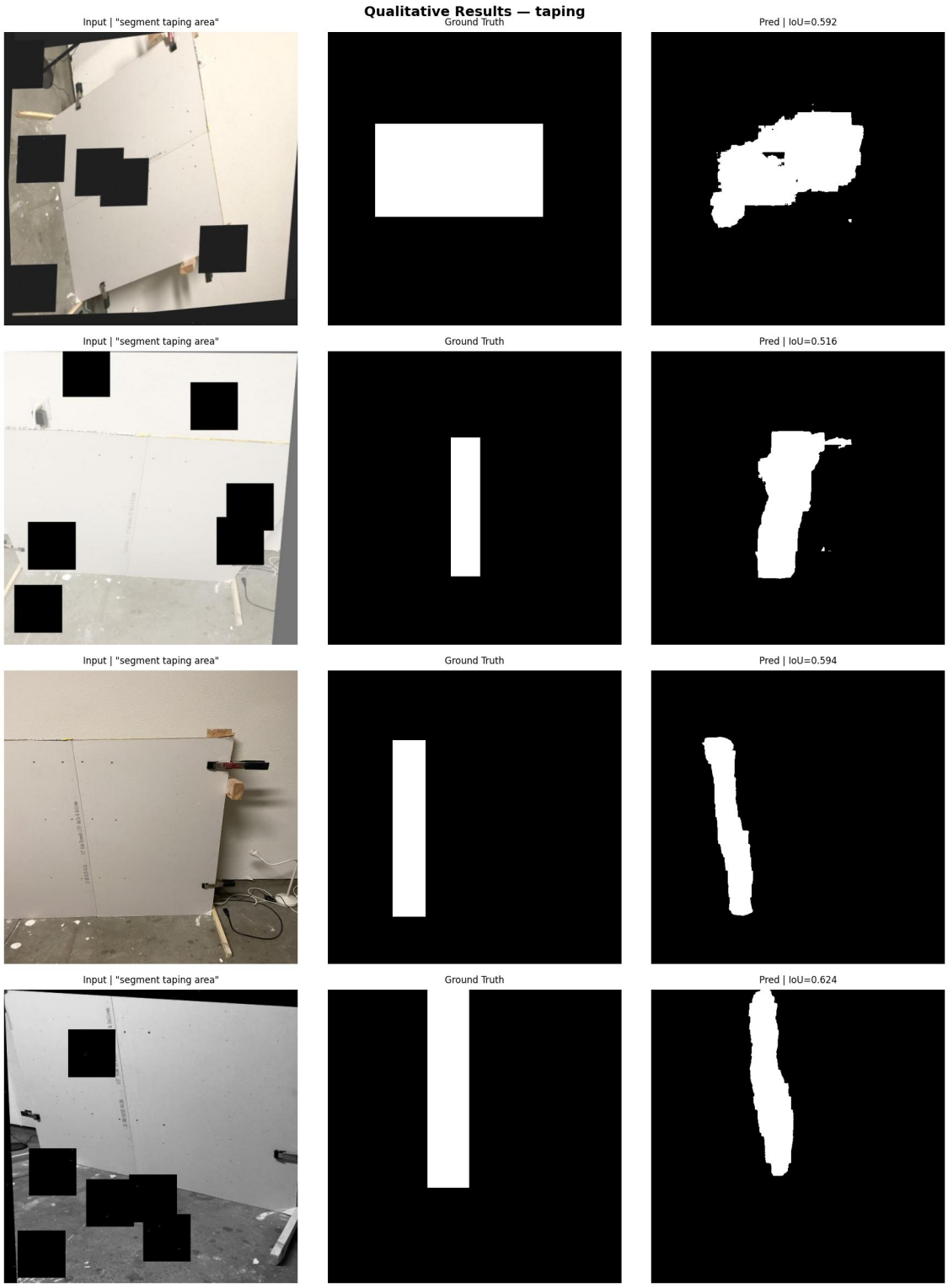

Figure 2: Qualitative results for "segment taping area". Model correctly localises vertical joint seams (IoU: 0.52–0.62).

Qualitative Results — Cracks

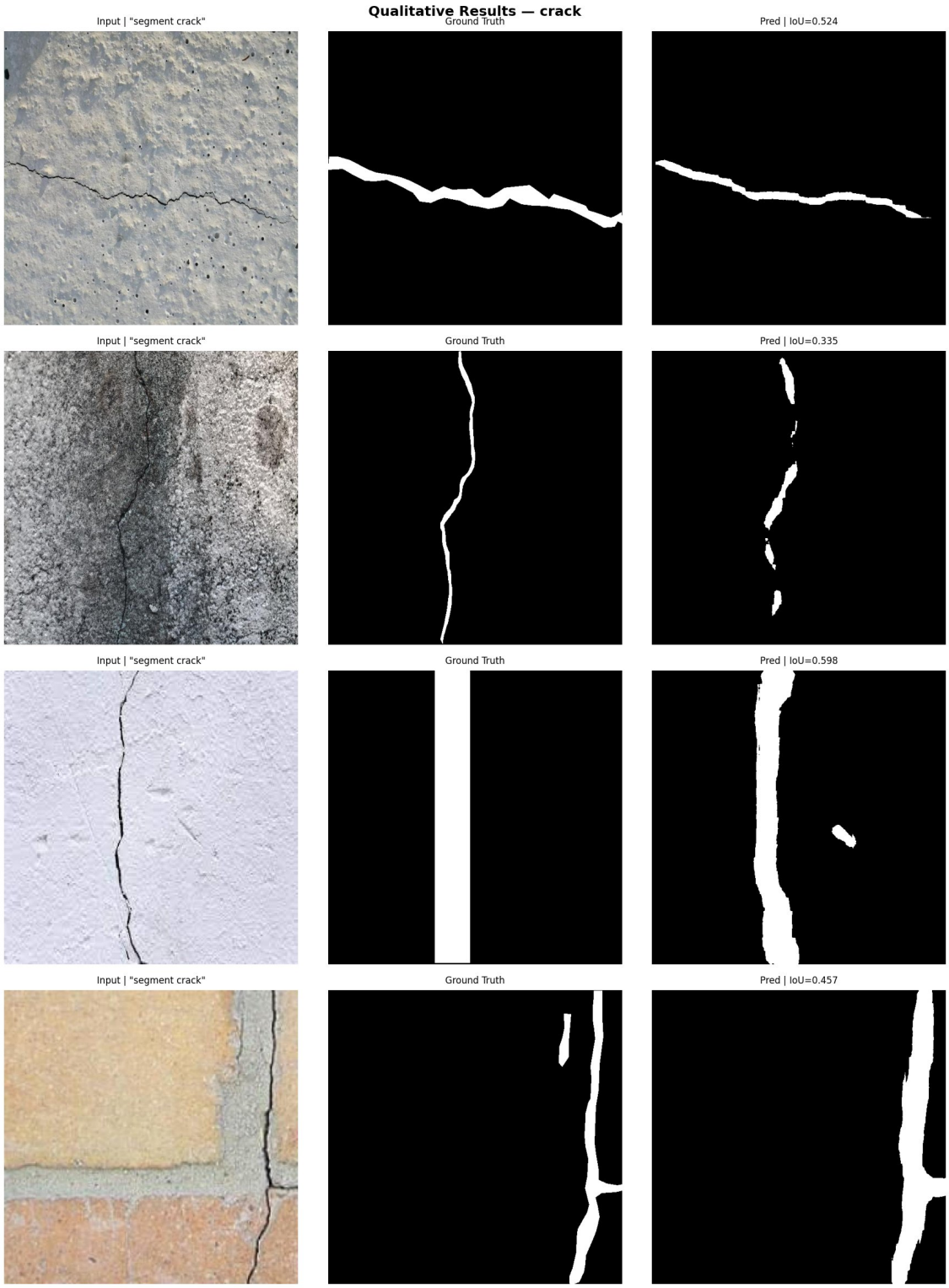

Figure 3: Qualitative results for "segment crack". Wide cracks well-detected (IoU: 0.52–0.60); fragmented predictions on thin/textured surfaces (IoU: 0.33–0.46).

4. Failure Cases & Potential Solutions

Analysis of low-IoU predictions from Figures 2 and 3 reveals four recurring failure modes:

Failure 1: Thin Hairline Crack Missed (Figure 3, Row 2 — IoU 0.335)

Cause: CLIPSeg uses 64px patch tokens. Sub-pixel hairline fractures fall within a single patch and receive low decoder activation — the spatial resolution is insufficient to resolve very thin linear structures.

Potential fixes:

- Multi-scale inference: run at 2–3 input scales, merge masks via max-pooling.
- CLAHE contrast enhancement and random sharpness augmentation during training.
- Fine-tune deeper ViT layers to reduce effective patch stride.

Failure 2: Fragmented Mask on Noisy Texture (Figure 3, Row 2 — IoU 0.335)

Cause: Coarse aggregate texture in the background (dark speckled concrete) activates the decoder locally, producing disconnected blob predictions around the true crack rather than a continuous mask.

Potential fixes:

- Post-processing: connected-component analysis, retain only the N largest components.
- Morphological closing (cv2.morphologyEx) to bridge prediction gaps.
- Add hard-negative training examples: similar texture regions with no crack annotation.

Failure 3: Bbox GT Mismatch on Taping (Figure 2, Row 1 — IoU 0.592)

Cause: Ground truth masks were generated from bounding box annotations. The GT rectangle is significantly larger than the actual tape seam, so the model is penalised for predicting a tighter, more accurate shape — artificially deflating IoU.

Potential fixes:

- Re-annotate with tight polygon masks using Roboflow's labelling tool.
- Use instance segmentation export format if available in the source dataset.
- Evaluate with boundary-aware IoU to reduce sensitivity to annotation looseness.

Failure 4: Low-Contrast Crack on Uniform Wall (Figure 3, Row 3 — IoU 0.598)

Cause: Thin cracks on white/light-coloured walls produce a very small intensity gradient at 352px resolution. The CLIP vision encoder struggles to distinguish the crack from the surrounding wall texture.

Potential fixes:

- Edge-aware preprocessing: concatenate a Sobel/Canny edge map as an auxiliary input channel.
- Semi-supervised pseudo-labeling on unlabeled construction site images to expand domain diversity.
- Random JPEG compression augmentation to simulate real site camera quality.

Appendix — Reproducibility Notes

- **Global seed:** 42 — applied to Python random, NumPy, PyTorch, and CUDA in Cell 0 of the notebook.
- **cuDNN flags:** deterministic=True, benchmark=False for full run-to-run reproducibility.
- **Environment:** Google Colab T4 GPU, Python 3.10, PyTorch 2.x, transformers==4.40.0.
- **Checkpoint selection:** Best model saved by average val mIoU across both classes per epoch.
- **Inference threshold:** 0.5 (default). A threshold sweep (0.3 / 0.5 / 0.7) is available in Cell 6b of the notebook.
- **Output format:** Single-channel PNG, values {0, 255}, same spatial resolution as input. Filename convention: *{image_id}__segment_crack.png* / *{image_id}__segment_taping_area.png*.

— Pranav Prashant Shewale —